AP-S/URSI — Technical Workshop HD-7
AI in Antenna Design and Analysis: From Pretrained Models to System-Level Integration

Part 2 — Pattern Reconstruction with AI

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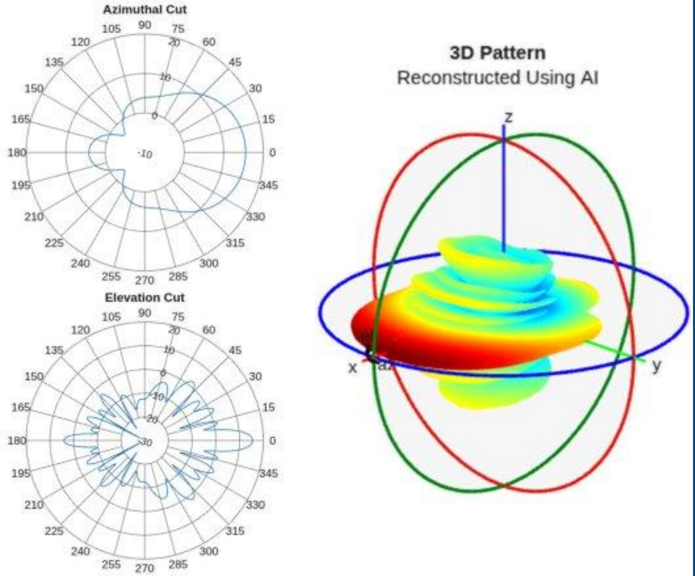
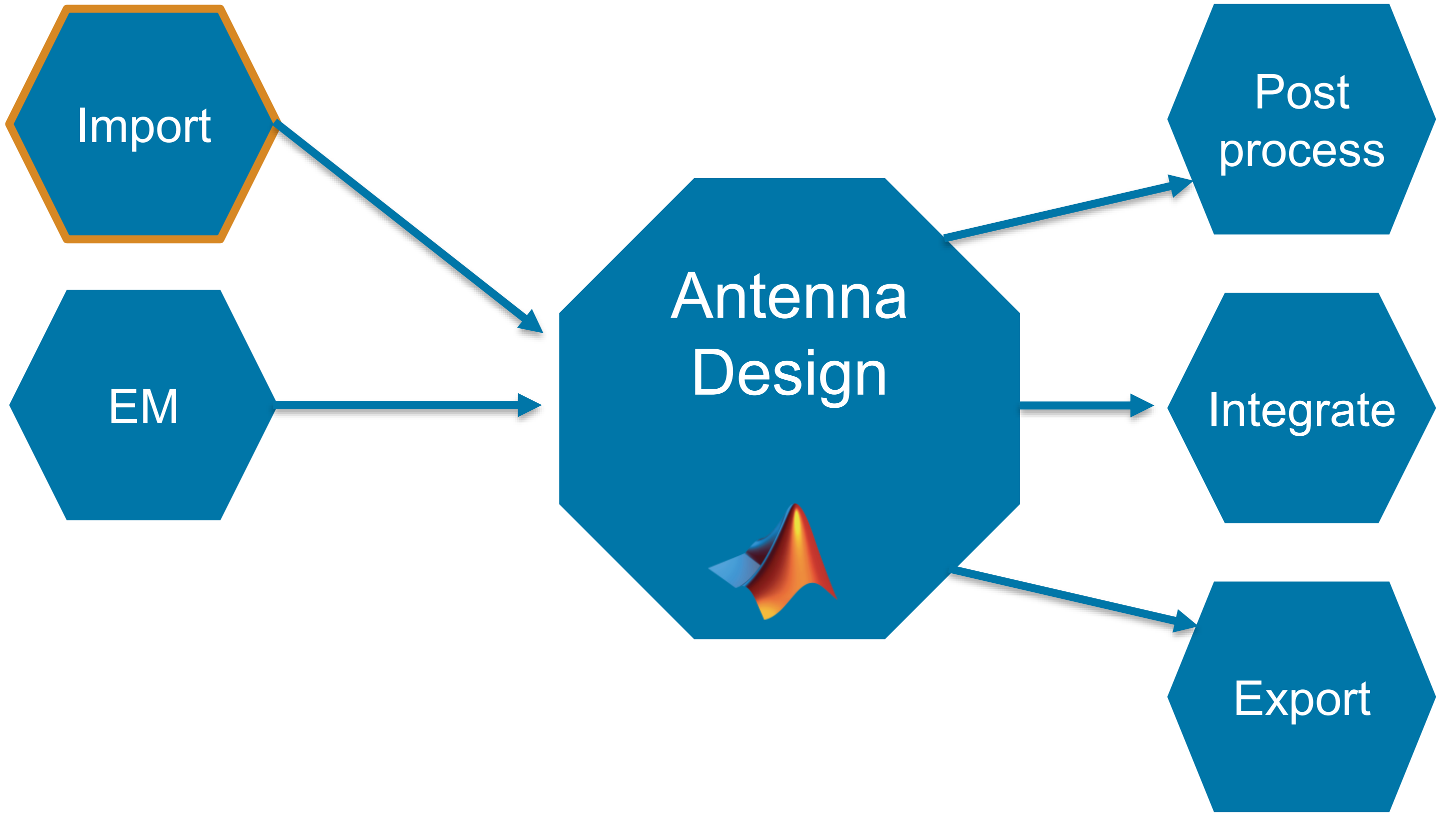
MathWorks | RF/EM Group



Antenna Workflows #2: Importing Data



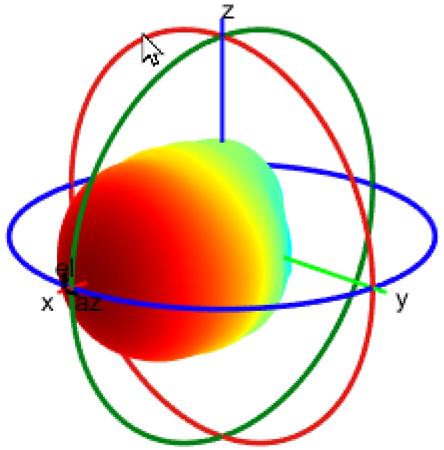
Importing Patterns – File Readers

- File Formats
  - .CSV
  - .MSI, .PLN
  - .FFS (CST)
  - .FFD (Ansys)
- Query on struct/object
- Visualize using
  - **`polarpattern`**
  - **`patternCustom`**

```
>> [Horizontal,Vertical,Optional] = msiread('Sample.pln')

Horizontal = struct with fields:
    PhysicalQuantity: 'Gain'
           Magnitude: [360x1 double]
               Units: 'dBd'
             Azimuth: [360x1 double]
           Elevation: 0
           Frequency: 659000000
               Slice: 'Elevation'

Vertical = struct with fields:
    PhysicalQuantity: 'Gain'
           Magnitude: [360x1 double
               Units: 'dBd'
             Azimuth: 0
           Elevation: [360x1 double
           Frequency: 659000000
               Slice: 'Azimuth'

Optional = struct with fields:
            name: 'Sample.pln'
            make: 'Sample 4DR-16-2HW'
       frequency: 659000000
         h_width: 180
         v_width: 7.3000
   front_to_back: 34
            gain: [1x1 struct]
            tilt: 'MECHANICAL'
    polarization: 'POL_H'
         comment: 'Ch-45 0 deg dt'
    scaling_mode: 'AUTOMATIC'
```

Command Window

```
>> data = ffsread("C:\temp\test2.ffs")
data =

  struct with fields:

             Version: "3.0"
              Format: "Farfield"
      NumFrequencies: 1
           Alignment: [1×1 struct]
       RadiatedPower: 0.4863
       AcceptedPower: 0.4969
     StimulatedPower: 0.5000
           Frequency: 3.3000e+11
                 Phi: [65341×1 double]
               Theta: [65341×1 double]
         EFieldTheta: [65341×1 double]
           EFieldPhi: [65341×1 double]
```



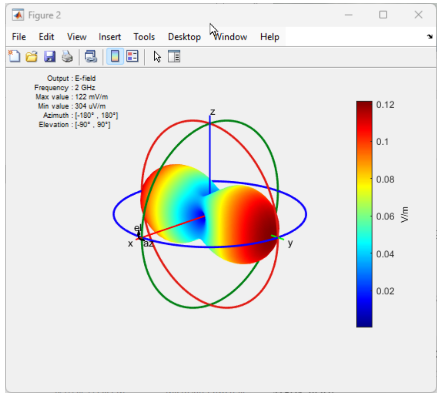
Using Antenna Data: Introducing the Measured Antenna Object

- Import antenna data: from measurements or 3rd party EM tools
  - S-parameters
  - E-field of (embedded) element

```
>> pattern(mesAnt,fRange(2),Type="efield")
>> EHfields(mesAnt,fRange(2))
>> sp = sparameters(mesAnt,fRange);
```

```
mesAnt = measuredAntenna(E=eField, Direction=dir, NumPorts=numAnt,...
            Azimuth=az, Elevation=el, FieldCoordinate="polar",...
            EmbeddedE=embE, FieldFrequency=fRange, Sparameters=s)
```

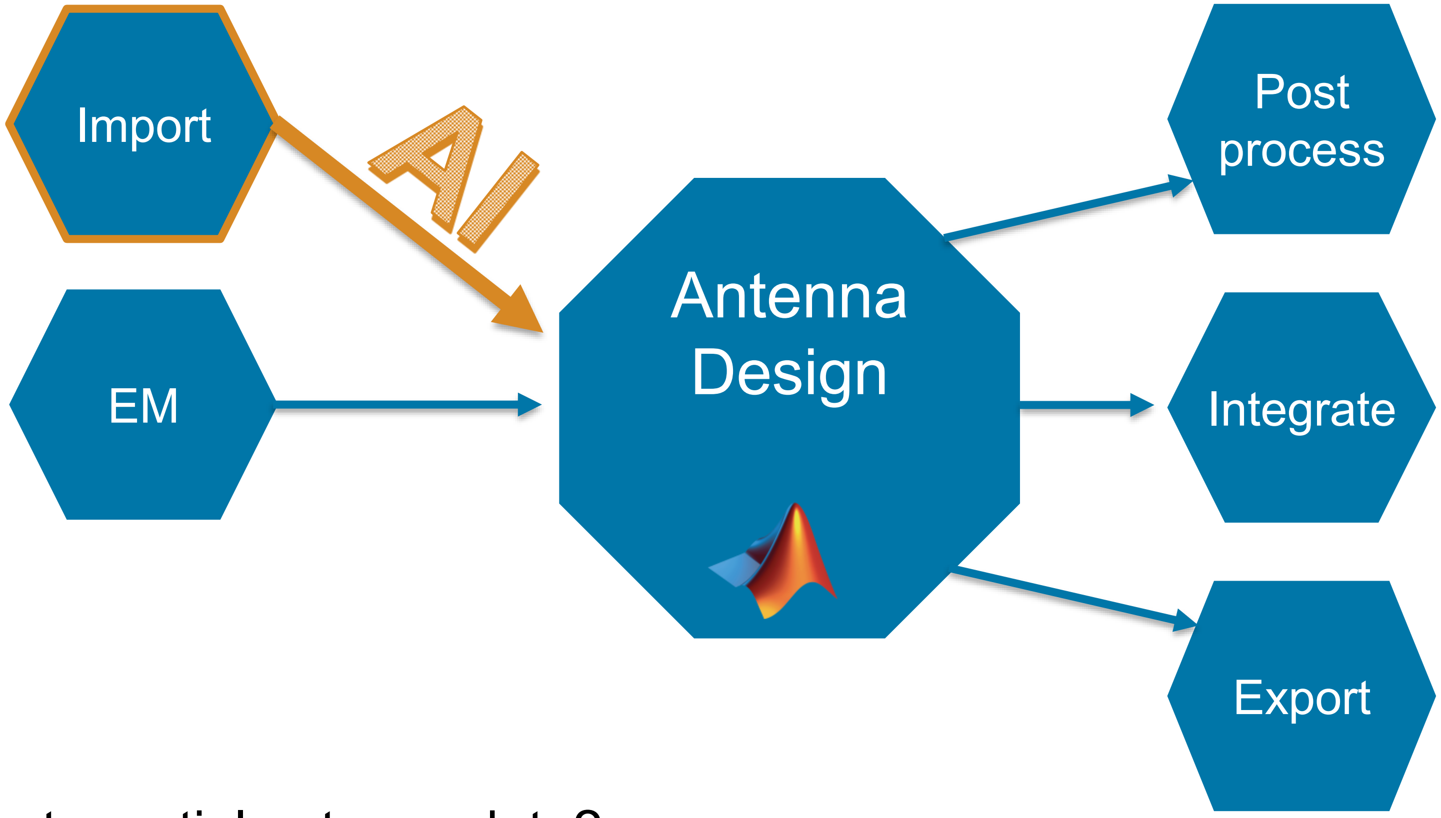
```
mesAnt =
  measuredAntenna with properties:

                    E: [2701x3x3 double]
            Direction: [2701x3 double]
          PhaseCenter: [0 0 0.0750]
             NumPorts: 2
       FieldFrequency: [3x1 double]
      FieldCoordinate: "polar"
              Azimuth: [-180 -175 -170 -165 -160 -155 -150
            Elevation: [180 175 170 165 160 155 150 145 14
          Sparameters: [1x1 sparameters]
        AmplitudeTaper: 1
           PhaseShift: 0
            EmbeddedE: [2701x3x2x3 double]
  TerminationImpedance: 50
    CalculateTotalField: 0
```



`Direction`: No. of observation points (Px3)
`FieldFrequency`: No. of frequency points (F)
`NumPorts`: Number of feeds (N)

`E`: P x 3 x F
`EmbeddedE`: P x 3 x N x F
`Sparameters`: N-port s-parameters object

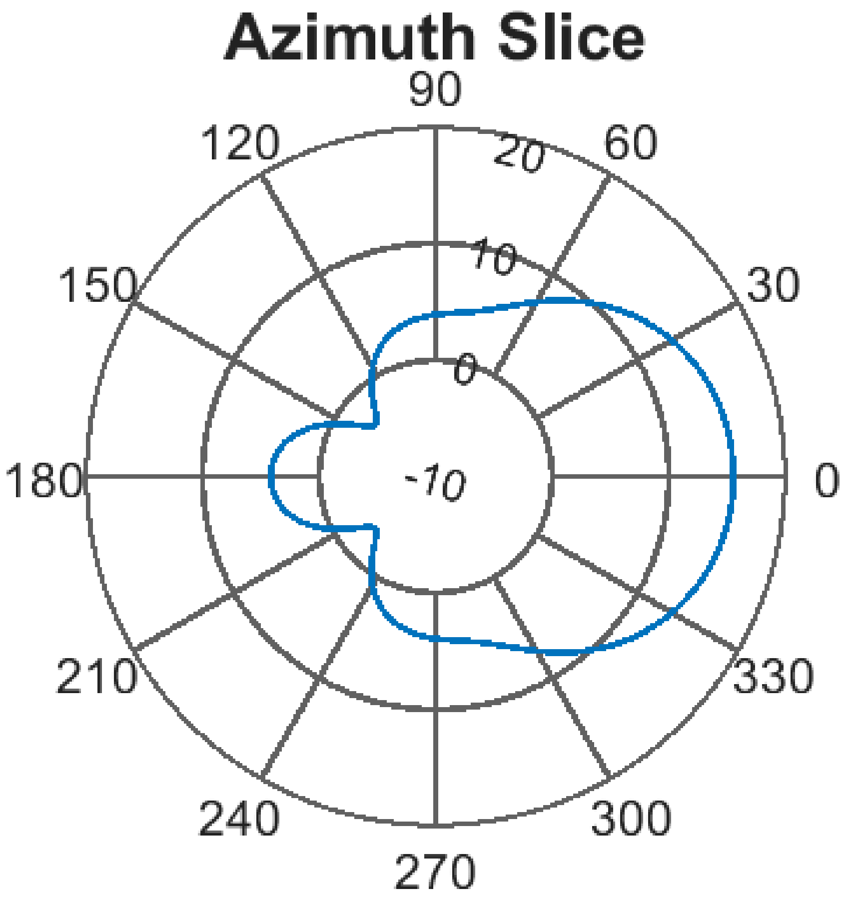
Antenna Workflows #2: Importing Data with AI

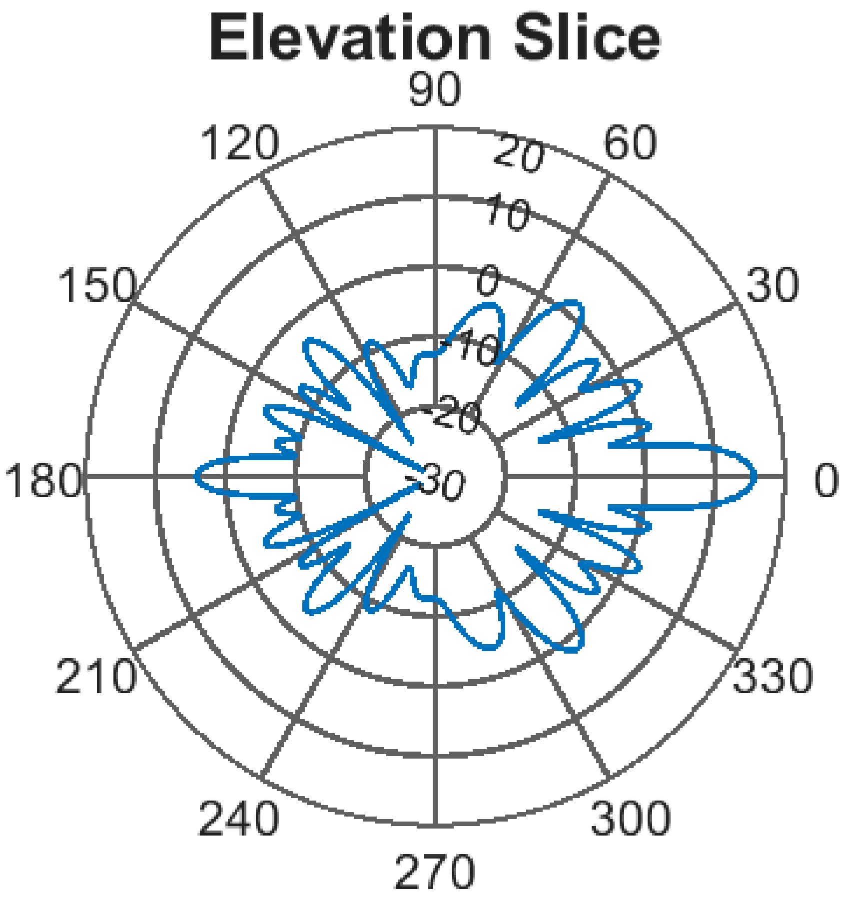


- Do you have access to partial antenna data?
- Do you need to integrate antennas at the system-level?
- Do you need to speed up antenna measurements?

What if my Pattern is not Complete?

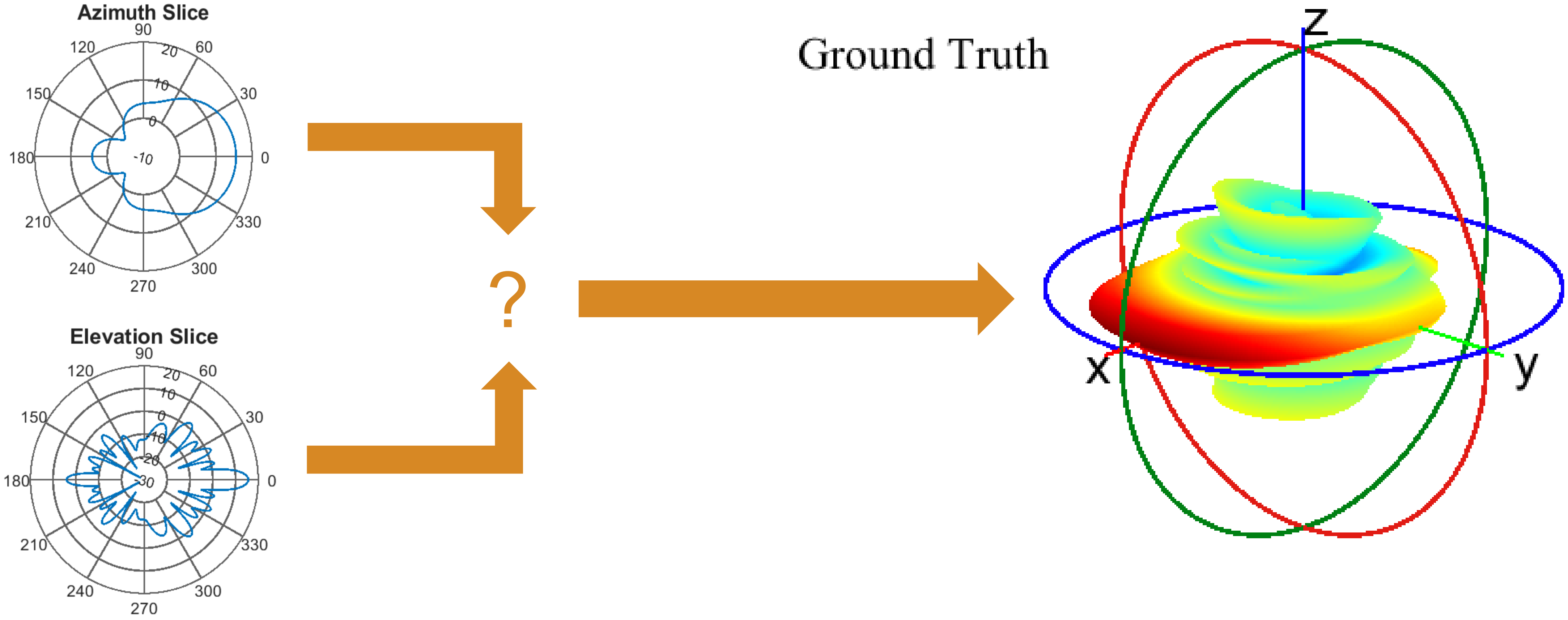
Can you reconstruct the full (3D) pattern?





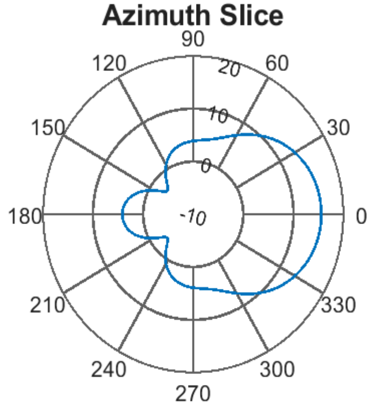
What if my Pattern is not Complete?

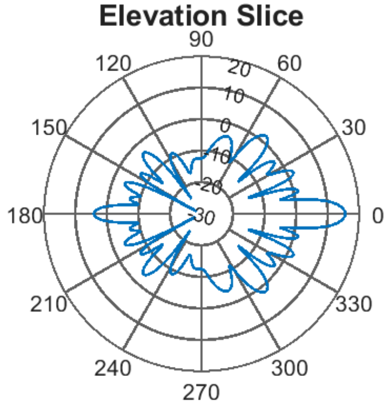
- Motivation for reconstructing 3D radiation patterns:
  - Enable system designers working with antenna datasheets (partial data)
  - Reduce the number of measurement samples



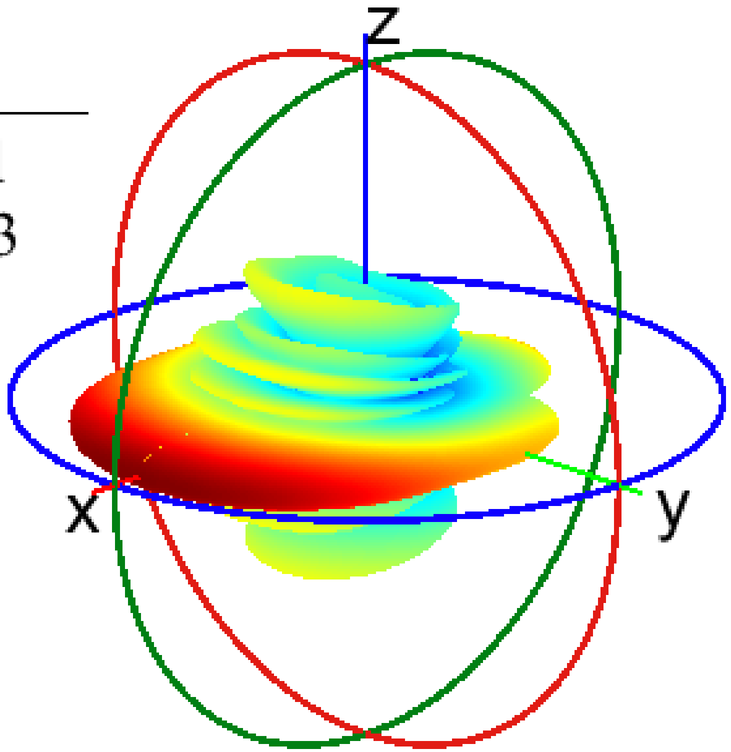
What if my Pattern is not Complete? 3D Pattern Reconstruction

- Reconstruct patterns starting from 2 (orthogonal) slices:
  - Analytical methods: **`patternFromSlices`**
  - Deep learning based prediction: **`patternFromAI`**





AI-Based
RMSE = 7.31
SSIM = 0.653



https://www.mathworks.com/help/antenna/ref/patternfromai.html
https://www.mathworks.com/help/antenna/ref/patternfromslices.html

AI-Enabled Pattern Reconstruction – *Overview*

Why AI?

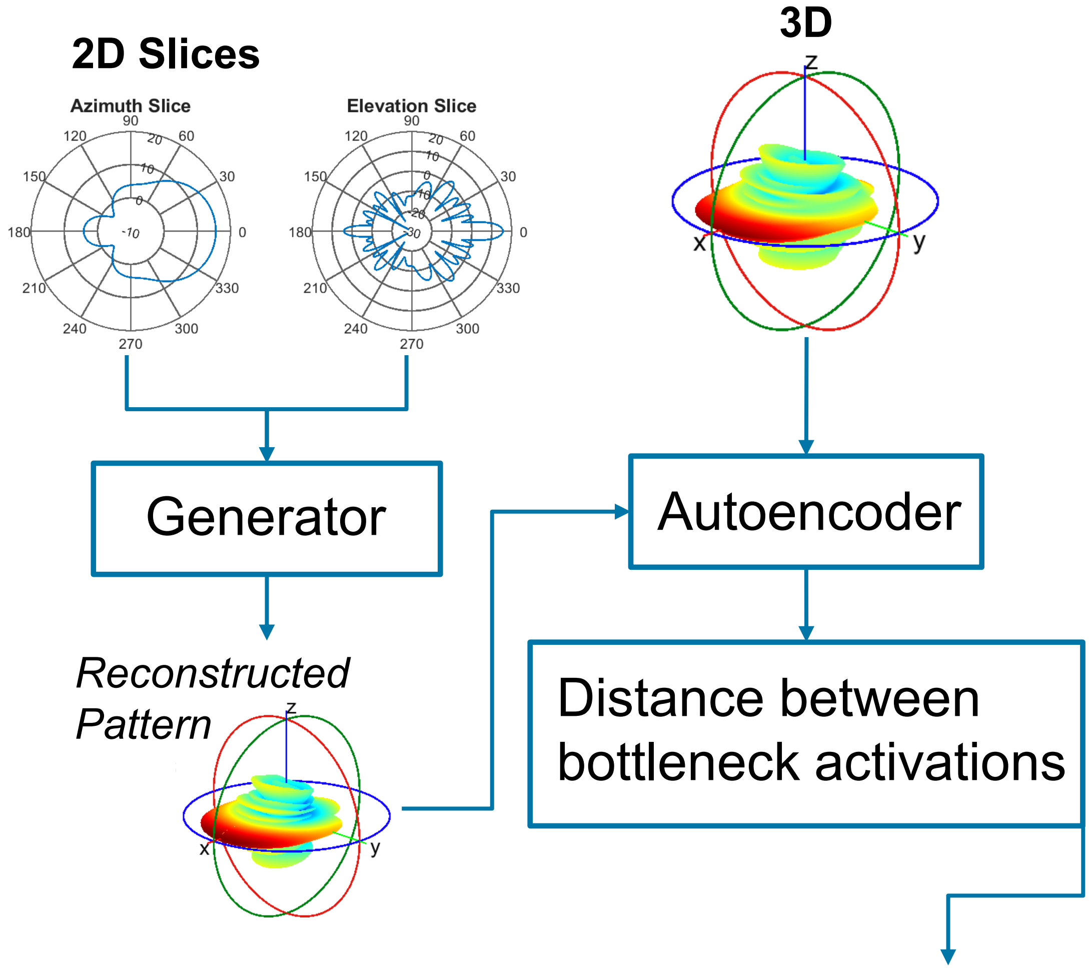
- Improve the accuracy of the reconstructed 3D pattern
- Overcome limitations of analytical approaches with no assumptions on input data format

Training Process

- Generator learns to predict 3D pattern given slices
- Custom loss function to compare against ground truth
  - L1: mean absolute error
  - SSIM: structural similarity
  - Feature-based loss

Training data: ~100,000 unique patterns (EM simulation)

- Single port antennas: omni / semi / highly-directional
- Array configuration: linear, circular, rectangular
- Classic image augmentations: noise, transformations



$$Total\ Loss\ =\ \alpha\cdot(L1\ Loss)+\beta\cdot(SSIM\ Loss)+\gamma\cdot(Feature\ Loss)$$

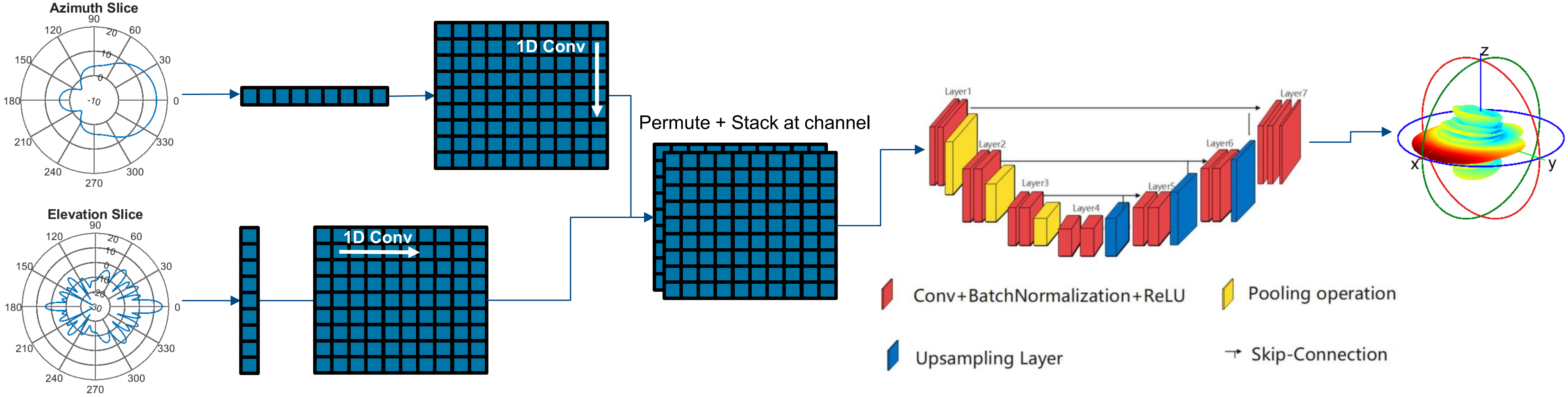
α,β,γ = hyper parameter weights

P. J. DiMeo, S. Sivaramakrishnan and V. Iyer, "AI-Based 3D Antenna Pattern Reconstruction," *2024 IEEE International Symposium on Antennas and Propagation and INC/USNC-URSI Radio Science Meeting (AP-S/INC-USNC-URSI)*, Firenze, Italy, 2024, pp. 1-2, doi: 10.1109/AP-S/INC-USNC-URSI52054.2024.10686954.

AI-Enabled Pattern Reconstruction – *Generator Architecture*

Generative model similar to multi-sequence-to-image convolutional neural network (CNN) with a U-Net backbone [1]

Construct sequence | **Expand 1D data to 2D** | **Generate Pattern Via UNet**

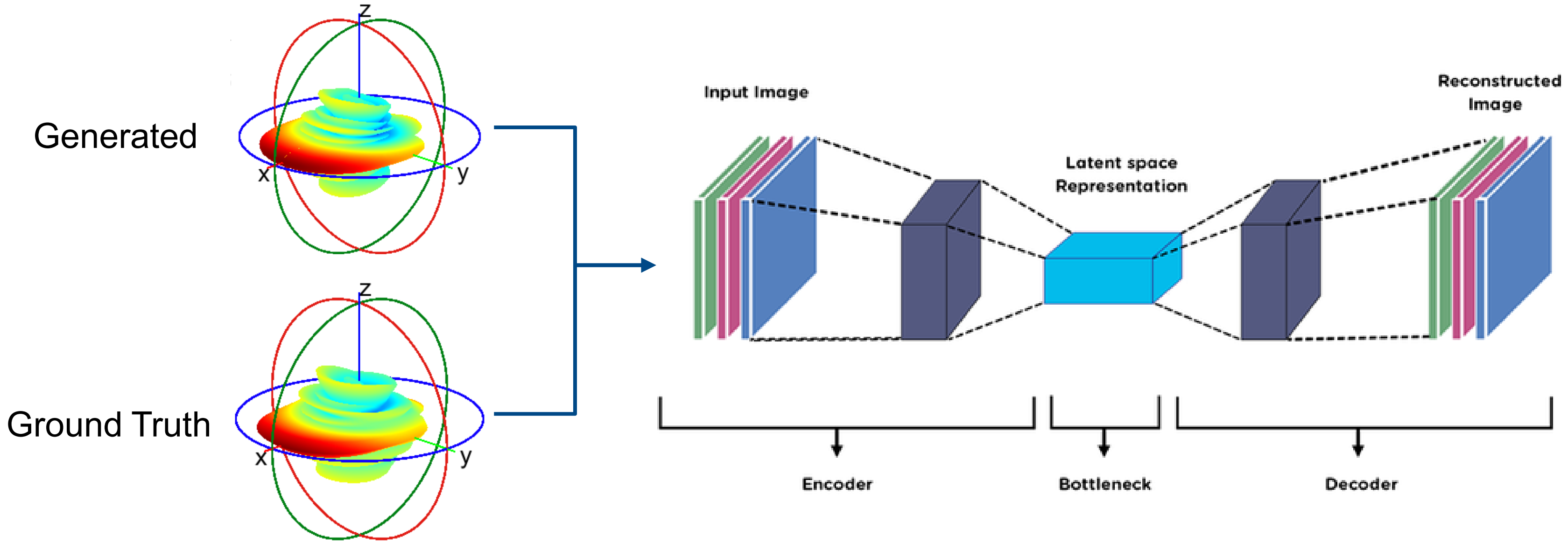


[1] V. Ghodrati, J. Shao, M. Bydder, Z. Zhou, W. Yin, K. L. Nguyen, Y. Yang, and P. Hu, "MR image reconstruction using deep learning: evaluation of network structure and loss functions," *Quant Imaging Med Surg.* vol. 9, no. 9, pp. 1516-1527, Sep. 2019. doi: 10.21037/qims.2019.08.10.
[2] S. Cai, Y. Wu, and G. Chen, "A Novel Elastomeric UNet for Medical Image Segmentation," *Front. Aging Neurosci.*, vol. 14, p. 841297, Mar. 2022. [Online]. Available: https://doi.org/10.3389/fnagi.2022.841297

AI-Enabled Pattern Reconstruction – *Feature Loss Autoencoder*

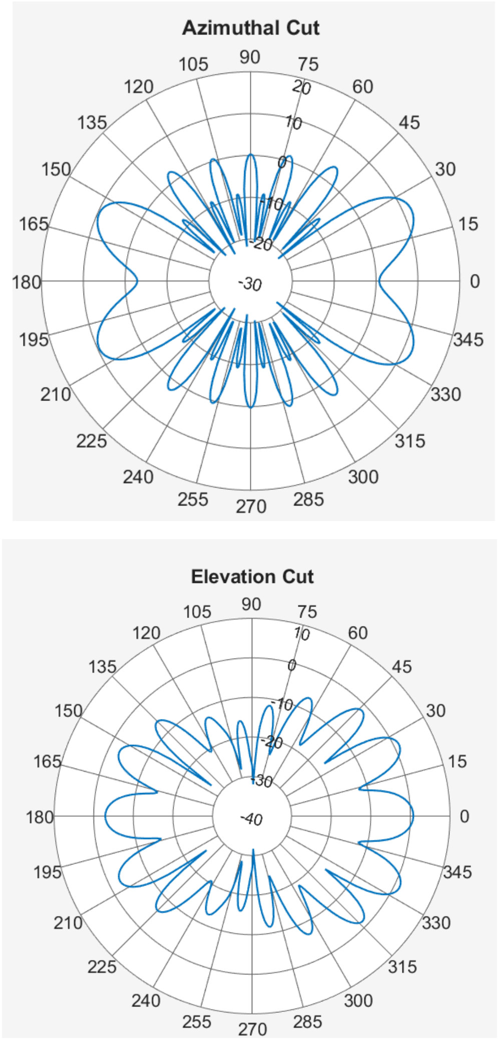
Custom-trained auto-encoder for (perceptual) feature-based loss

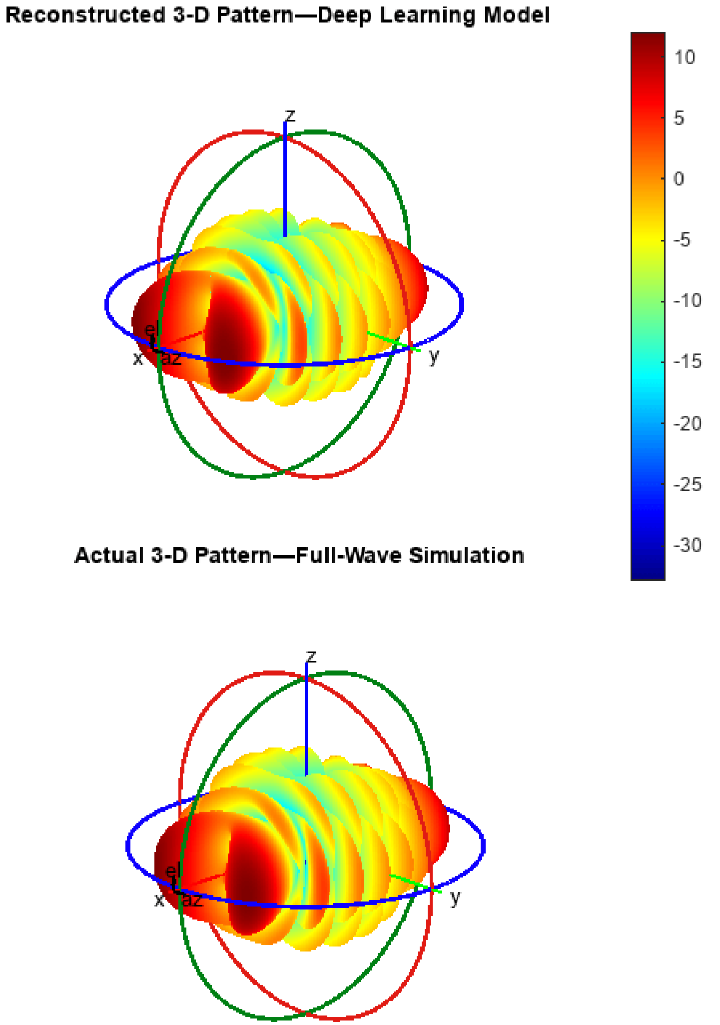
- Perceptual loss captures antenna specific features
- Bottleneck activations serve as the semantic representation
- Comparing activations at the **bottleneck** will guide us toward a more perceptually similar generated pattern.

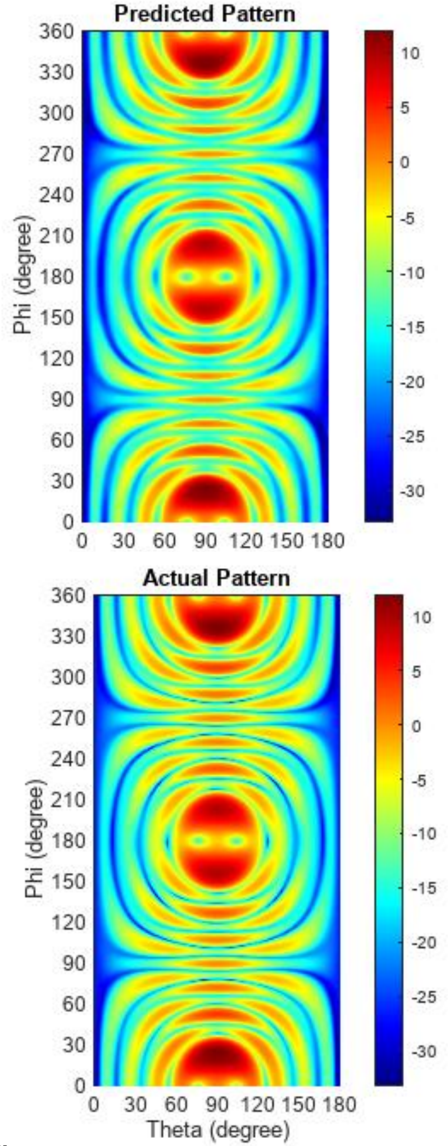


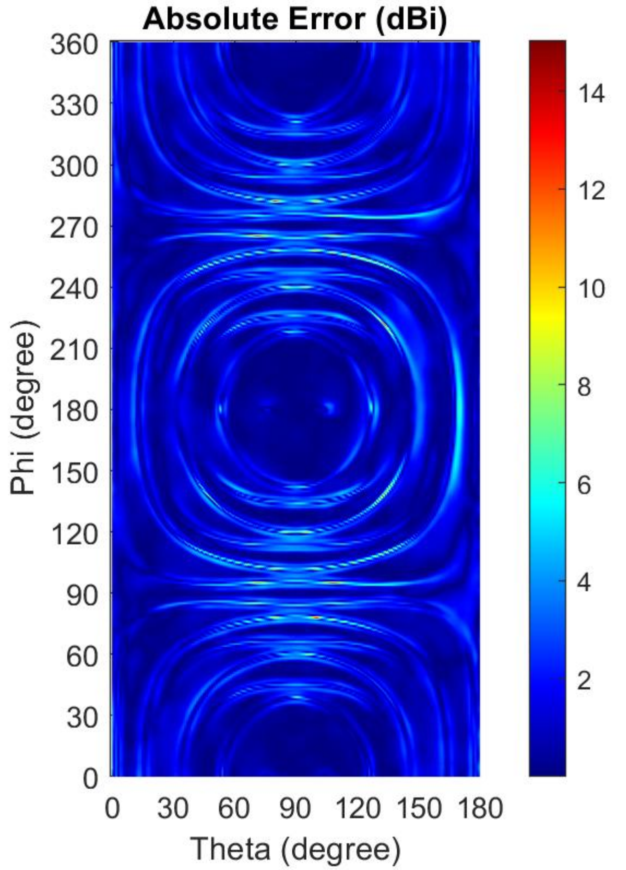
[1] https://medium.com/@birla.deepak26/autoencoders-76bb49ae6a8f

Pattern Reconstruction Using AI: *Simulation Data Results*









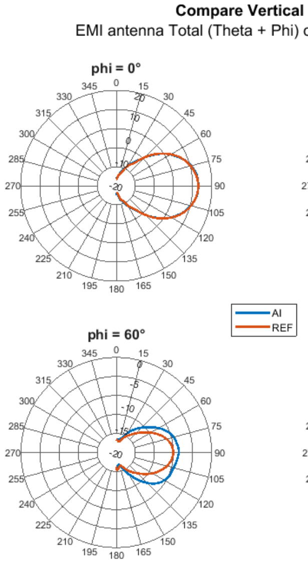
Reconstruct 3-D Antenna Pattern from 2-D Slices Using Deep Learning

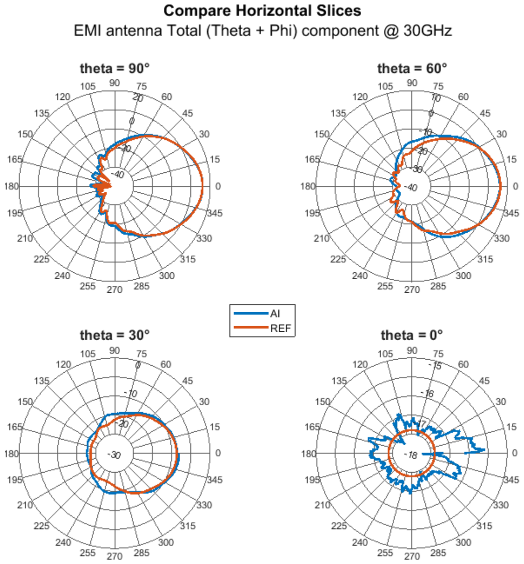
https://www.mathworks.com/help/antenna/ug/reconstruct-3d-pattern-from-2d-slices-using-deep-learning.html

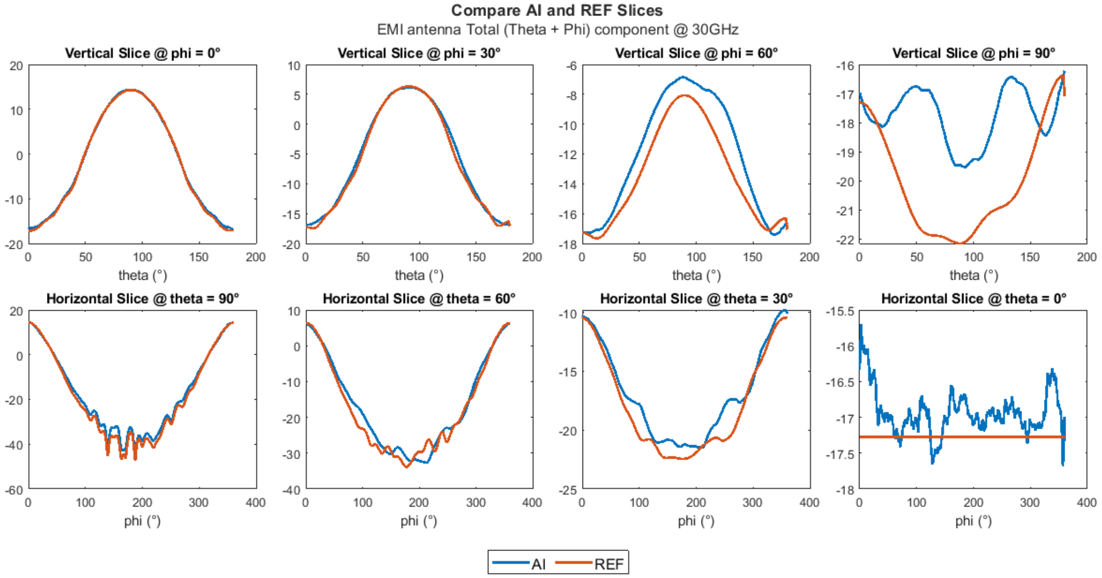
Pattern Reconstruction Using AI: *Measured Data Results*

Collaboration with Rohde&Schwarz

- Unique commercial AI-based model to reconstruct diverse antenna pattern types
- Reconstruct 3D pattern from sparse data (~ **1%** of full pattern data) with low RMSE and high SSIM
- Small model size – **30 MB**
- Fast inference times of ~ <= **1sec**







Rohde&Schwarz Test Forum video: [Accelerating antenna pattern analysis and simulation with MATLAB]

With special thanks and acknowledgements to the work of Adam Tankielun (Rohde & Schwarz)

Interactive Exercises Using MATLAB Online R2026a

Live Demo Using R2026b Prerelease

Review and 3p integration workflow:

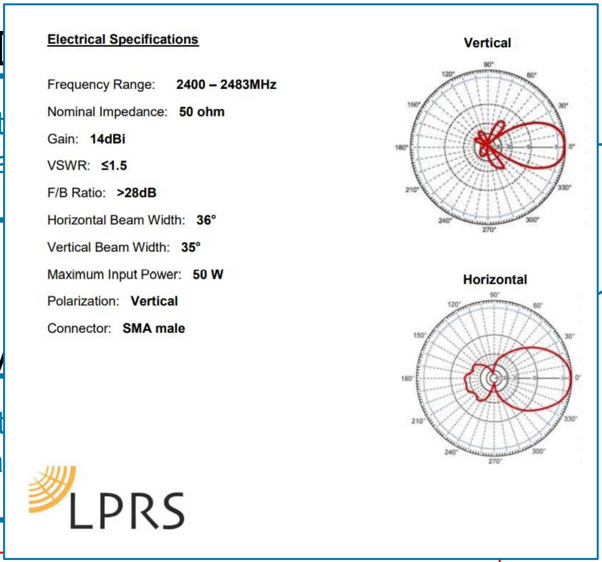
Convert image into 2D slice data (26a)

D

2d pat
S-para

M

2d patt
S-para

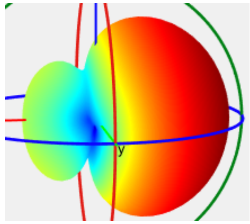

Electrical Specifications

Frequency Range: 2400 – 2483MHz
Nominal Impedance: 50 ohm
Gain: 14dBi
VSWR: ≤1.5
F/B Ratio: >28dB
Horizontal Beam Width: 36°
Vertical Beam Width: 35°
Maximum Input Power: 50 W
Polarization: Vertical
Connector: SMA male

Read 3p file formats : ffs (25b)

3D pattern reconstruction

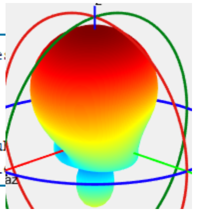
patternFromAI



3p simulations

3d complex pattern data
S-parameter touchstone file

```
sparameters with propertie
    Impedance: 50
     NumPorts: 1
   Parameters: [1×1×31 dou
  Frequencies: [31×1 doubl
```



Measurement

3d complex pattern data
S-parameter touchstone file

Measured Antenna

```
measuredAntenna with properties:

               E: [190×3×3 double]
       Direction: [190×3 double]
     PhaseCenter: [0 0 0]
        NumPorts: 1
  FieldFrequency: [3×1 double]
 FieldCoordinate: 'polar'
         Azimuth: [-180 -160 -140 -12
       Elevation: [90 70 50 30 10 -10
     Sparameters: [1×1 sparameters]
```

System Simulation

PhasedArray

Ray Tracing

Satcom

Comms
Standards

Radar